Exercise 1 :

Calculate $P(A)$, $P(B)$, $P(A|B)$ from the joint probability distribution $P(A, B)$ given in Table 1.

	b_1	b_2	b_3
a_1	0.05	0.10	0.05
a_2	0.15	0.00	0.25
a_3	0.10	0.20	0.10

Table 1: The joint probability distribution for $P(A, B)$.

Solution: In order to find $P(A)$ we need to marginalize (sum) out the variable B. This is done for each state of A. Thus, for $A = a_1$ we get:

$$P(A = a_1) = \sum_B P(A = a_1, B) = 0.05 + 0.10 + 0.05 = 0.20$$

In total we end up with $P(A) = (0.2, 0.4, 0.4)$. Using a similar procedure to find $P(B)$ (now summing out A) we get $P(B) = (0.3, 0.3, 0.4)$.

In order to find $P(A|B)$ we use

$$P(A|B) = \frac{P(A, B)}{P(B)},$$

hence for $B = b_1$ we get

$$P(A|b_1) = \frac{P(A, B = b_1)}{P(B = b_1)} = \frac{(0.05, 0.15, 0.10)}{0.3} = (0.167, 0.5, 0.333).$$

The operations are similar for the other states of B, which gives $P(A|b_2) = (0.333, 0, 0.667)$ and $P(A|b_3) = 0.125, 0.625, 0.25)$.

Last, if we want to compute also $P(B|A)$ (not asked in the exercise), we can calculate these probabilities following the same steps as for $P(A|B)$, resulting in

$$P(B|a_1) = (0.25, 0.5, 0.25)$$
$$P(B|a_2) = (0.375, 0, 0.625)$$
$$P(B|a_3) = (0.25, 0.5, 0.25).$$

Exercise 2 :

Consider the network in Figure 1. Which conditional probability tables must be specified to turn the graph into a Bayesian network? Assuming that all variables are binary, what is the size of those tables (i.e., how many values do we need to specify?)

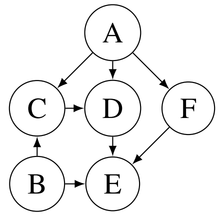
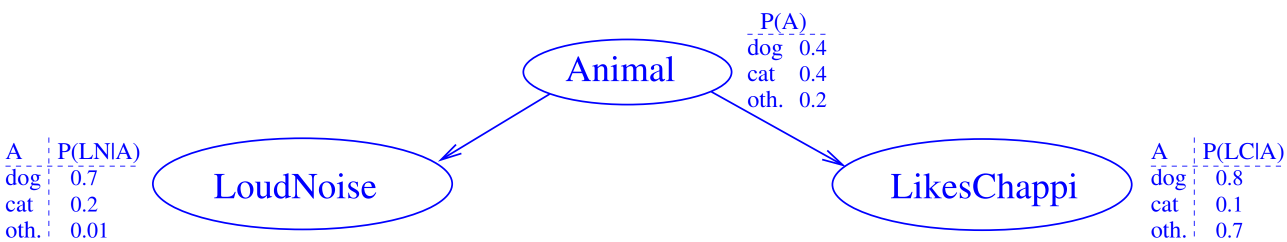


Figure 1: A Bayesian network for Exercise 2.

Solution: The needed tables are $P(A)$, $P(B)$, $P(C \mid A, B)$, $P(D \mid A, C)$, $P(E \mid B, D, F)$, and $P(F \mid A)$. The sizes are $2 + 2 + 8 + 8 + 16 + 4 = 40$.

Exercise 3 :

Consider the following Bayesian network BN:



Use this network to compute the following probabilities:

(a) $P(loudnoise, dog, likeschappi)$.

(b) $P(loudnoise, other, likeschappi)$.

(c) $P(\neg loudnoise, dog, likeschappi)$.

Include intermediate steps at a level of granularity as in the lecture slides examples. In particular, for each of the probabilities P demanded in (a) – (c), write down *which* probabilities provided in BN can be combined *how* to obtain P.

Solution:

In all cases, we use the method from slides 42 and 43. As the variable ordering consistent with BN, we choose $X_1 = Animal$, $X_2 = LoudNoise$, $X_3 = LikesChappi$.

(a) $P(loudnoise, dog, likeschappi) = P(likeschappi \mid loudnoise, dog) * P(loudnoise \mid dog) * P(dog) = P(likeschappi \mid dog) * P(loudnoise \mid dog) * P(dog) = 0.8 * 0.7 * 0.4 = 0.224$.

(b) $P(loudnoise, other, likeschappi) = P(likeschappi \mid loudnoise, other) * P(loudnoise \mid other) * P(other) = P(likeschappi \mid other) * P(loudnoise \mid other) * P(other) = 0.7 * 0.01 * 0.2 = 0.0014$.

(c) $P(\neg loudnoise, dog, likeschappi) = P(likeschappi \mid \neg loudnoise, dog) * P(\neg loudnoise \mid dog) * P(dog) = P(likeschappi \mid dog) * P(\neg loudnoise \mid dog) * P(dog) = 0.8 * 0.3 * 0.4 = 0.096$. Note here that $P(\neg loudnoise \mid dog)$ is not explicitly given in the picture, but can be calculated as $1 - P(loudnoise \mid dog)$; compare Chapter 16 slide 12.

Exercise 4 :

We want to construct a Bayesian network for the following random variables (all with domain *true*,*false*):

Mexico	Person X has recently travelled to Mexico
Svine_flu_infection	Person X has been infected with the svine flu virus
Vaccination	Person X has been vaccinated against svine flu
Svine_flu_sick	Person X is sick from svine flu
Fever	Person X has fever

a. Use the above ordering of the variables to determine a Baysian network structure based on the chain rule and conditional independence relations.

b. Repeat the construction with the alternative variable ordering:

Vaccination,*Mexico*,*Svine_flu_sick*,*Fever*,*Svine_flu_infection*

Solution:

a. We write the joint distribution of the variables using the chain rule:

$$\begin{aligned} & P(M, S\_f\_i, V, S\_f\_s, F) = \\ & \quad P(M)\cdot \\ & \quad P(S\_f\_i \mid M)\cdot \\ & \quad P(V \mid M, S\_f\_i)\cdot \\ & \quad P(S\_f\_s \mid M, S\_f\_i, V)\cdot \\ & \quad P(F \mid M, S\_f\_i, V, S\_f\_s)\cdot) \end{aligned}$$

We now simplify the conditional distributions by making some conditional independence assumptions (these are reasonable assumptions based on our knowledge of the domain – but not necessarily provably correct).

$$P(V \mid M, S\_f\_i) = P(V \mid M)$$

Justification: whether or not a person gets a vaccination may depend on whether he plans to travel to mexico. His decision to get a vaccination (generally) takes place before he the potential time of infection, so the decision is independent of whether he will catch the virus.

$$P(S\_f\_s \mid M, S\_f\_i, V) = P(S\_f\_s \mid S\_f\_i, V)$$

Justification: Given that we know whether a person has been infected with the virus, and whether or not he has a vaccination, it is no longer relevant to know where he may have contracted the virus, especially whether or not he traveled to Mexico. Note that this does not mean that $P(S\_f\_s \mid M) = P(S\_f\_s)$: as long as we don't know about S_f_s and V, the information of whether the person traveled to Mexico is still very relevant for the probability of S_f_s.

$$P(F \mid M, S\_f\_i, V, S\_f\_s) = P(F \mid S\_f\_s)$$

Justification: the fever is a direct consequence of the sickness. Once we know whether a person is sick, it gives us no additional useful information (regarding the probability of fever) to know whether the person was in Mexico, or got a vaccination.

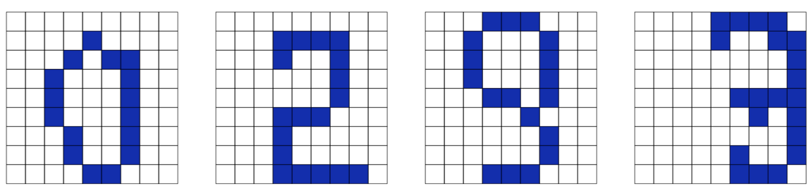
The representation of the joint distribution as the product of the simplified conditional distributions corresponds to the Bayesian network structure

b.

Going through the same procedure for the alternative ordering leads to fewer simplifications, and a more complicated network structure. The basic reason for this is that the second ordering does not respect the cause-effect relationships between the variables (i.e., taking cause variables before their effects).

Exercise 5 :

Design a Bayesian network that can be used to recognize handwritten digits 0,1,2,. . . ,9 from scanned, pixelated images like these:



- What are hypothesis and information variables?
- Could there be any useful mediating variables (consider e.g. the last image above)?
- How could you design a network structure
  - so that the conditional independencies are (approximately) reasonable
  - so that specification and inference complexity remain feasible
- How do you fill in the conditional probability tables?

Solution:

- The hypothesis variable(s) must enable us to make the intended inferences by querying these variables. Here we are interested in predicting which digit is actually represented by the image. Thus, we should have a variable

$$\textit{digit} \in \{0, 1, \ldots, 9\}$$

The information variables must allow us to enter into the Bayesian networks the relevant observations which we make. Here we observe $9 \cdot 9 = 81$ pixels, each of which can be black or white:

$$\textit{pixel\_i} \in \{b, w\} \quad i = 1, \ldots, 81$$

- Mediating variables: see below
- Writing a given digit can be seen as a cause for the pixels to become black or white. Just inserting edges for these direct cause-effect relationships gives us the structure According to this structure, any two pixels are conditionally independent given the digit (this type of structure is also called a *Naive Bayes Model*). This is not entirely realistic: for example consider pixels 27 and 36: According to the network structure:

$$P(\textit{pixel\_27} = b \mid \textit{digit} = 3, \textit{pixel\_36}{=}b) = P(\textit{pixel\_27} = b \mid \textit{digit} = 3),$$

i.e. knowing that the actual digit is a 3, the information on pixel 36 carries no relevant information for pixel 27. However, if in addition to *digit*=3 we know that *pixel_36*=b, this will indicate that the digit is written flush right in the box, and should thereby increase the probability of *pixel_27*=b.

We can improve the model by introducing a mediating variable

$$\textit{alignment} \in \{l, r, c\}$$

(for left, right, and center alignment) and add it to the network like this: Now two pixels are only independent given both *digit* and *alignment* – which still will not make this a perfectly accurate model, but already better than the first one.

Exercise 6 :

For 10000 emails in your inbox you determine the values of the following three boolean variables:

Spam	the email is spam
Caps	the subject line is in all capital letters
Pills	Body of the message contains the word “pills”

You obtain the following counts:

	Caps			
	yes		*no*	
	Pills		*Pills*	
Spam	*yes*	*no*	*yes*	*no*
yes	150	850	600	3400
no	1	99	49	4851

Are

- *Spam* and *Caps* independent?
- *Pills* and *Caps* independent?

- *Pills* and *Caps* independent given *Spam*?
- *Spam* and *Caps* independent given *Pills*?

Solution:

To determine whether *Spam* and *Caps* are independent, we consider the joint and marginal distribution of these two variables:

	Caps		
Spam	*yes*	*no*	
yes	0.1	0.4	0.5
no	0.01	0.49	0.5
	0.11	0.89	

The entries are obtained by summing over the *Pills* variable, and normalizing by dividing by 10000. E.g. $0.4 = \frac{600+3400}{10000}$. Since, for example $P(\textit{Spam} = \textit{no}, \textit{Caps} = \textit{yes}) = 0.01 \neq P(\textit{Spam} = \textit{no})P(\textit{Caps} = \textit{yes}) = 0.5 \cdot 0.11 = 0.055$, we see that the two variables are not independent.

To see whether *Spam* and *Caps* are independent given *Pills*, we first determine the conditional distribution of *Spam* and *Caps* given *Pills*=*yes*:

	Caps		
Spam	*yes*	*no*	
yes	0.1875	0.75	0.9375
no	0.00125	0.06125	0.0625
	0.18875	0.81125	

Here, e.g. $0.1875 = \frac{150}{800}$ (800 is the total number of cases with *Pills*=*yes*).

Again, we find that the joint distribution is not the product of the marginals, e.g. $0.0625 \cdot 0.18875 = 0.0118 \neq 0.00125$. Thus, *Spam* and *Caps* are not independent given *Pills*.

For the (conditional) independence of *Pills* and *Caps* one proceeds in the same manner. Now the findings should be: *Pills* and *Caps* are not independent, but *Pills* and *Caps* are independent given *Spam*. For the last result one has to check both the conditional distribution given *Spam*=*yes* and given *Spam*=*no* (above it was enough to consider the conditional distribution of *Spam* and *Caps* given *Pills*=*yes*, because there we already found that *Spam* and *Caps* are not conditionally independent).

Exercise 7 :

Consider the experiment of flipping a coin, and if it lands heads, rolling a four-sided die, and if it lands tails, rolling a six-sided die.

(i) Suppose that the coin and both dice are fair. As we are interested only in the number rolled by the die, and the possible worlds $\mathcal{S}_A$ for the experiment could thus be the numbers from 1 to 6. Another set of possible worlds could be $\mathcal{S}_B = \{t1, \ldots, t6, h1, \ldots, h4\}$, with for example $t2$ meaning "tails and a roll of 2" and $h4$ meaning "heads and a roll of 4." Choose either $\mathcal{S}_A$ or $\mathcal{S}_B$ and associate probabilities with it. According to your chosen set of possible worlds and probability distribution, what is the probability of rolling either 3 or 5.

(ii) Let $\mathcal{S}_B$ be defined as above, but with a loaded coin and loaded dice. A probability distribution is given in Table 2. What is the probability that the loaded coin lands "tails"? What is the conditional probability of rolling a 4, given that the coin lands tails? Which of the loaded dice has the highest chance of rolling 4 or more?

$t1$	$\frac{5}{18}$	$t6$	$\frac{1}{18}$
$t2$	$\frac{1}{9}$	$h1$	$\frac{1}{24}$
$t3$	$\frac{1}{9}$	$h2$	$\frac{1}{24}$
$t4$	$\frac{1}{18}$	$h3$	$\frac{1}{8}$
$t5$	$\frac{1}{18}$	$h4$	$\frac{1}{8}$

Table 2: Probabilities for $\mathcal{S}_B$.

Solution:

(i) Probabilities for $\mathcal{S}_{\mathcal{A}}$: $P_A(1) = \cdots = P_A(4) = \frac{5}{24}$ and $P_A(5) = P_A(6) = \frac{1}{12}$.
Probabilities for $\mathcal{S}_{\mathcal{B}}$: $P_B(t1) = \cdots = P_B(t6) = \frac{1}{12}$ and $P_B(h1) = \cdots = P_B(h4) = \frac{1}{8}$.
$P_A(3) + P_A(5) = \frac{7}{24}$.
$P_B(t3) + P_B(t5) + P_B(h3) = \frac{7}{24}$.

(ii) $P(t) = \frac{5}{18} + \frac{1}{9} + \frac{1}{9} + \frac{1}{18} + \frac{1}{18} + \frac{1}{18} = \frac{2}{3}$.
$P(4 \,|\, t) = P(t4)/P(t) = \frac{1}{12}$.
$P(4 \,|\, t) + P(5 \,|\, t) + P(6 \,|\, t) = \frac{1}{4}$.
$P(4 \,|\, h) = P(h4)/P(h) = 1/8/(\frac{1}{24} + \frac{1}{24} + \frac{1}{8} + \frac{1}{8}) = \frac{3}{8}$.
The four-sided die thus has a higher probability of rolling 4 or more, than the six-sided die.

Exercise 8 :

Consider the binary variable A and the ternary variable B. Assume that B has the probability distribution $P(B) = (0.1, 0.5, 0.4)$ and that A has the conditional probability distribution given in Table 3.

	b_1	b_2	b_3
a_1	0.1	0.7	0.6
a_2	0.9	0.3	0.4

Table 3: The conditional probability distribution $P(A|B)$.

Questions:

(i) Verify that Table 3 specifies a valid conditional probability distribution.

(ii) Calculate $P(B|A)$. *Hint:* Consider Bayes rule illustrated by the temperature-sensor example that we discussed in the lecture

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Solution:

1. We need to check that $\sum_A P(A|B = b_i) = 1$ for each state b_i of B. This is the case in Table 3, hence the we have a valid conditional probability distribution.

2. According to Bayes rule we have

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{P(A|B)P(B)}{\sum_B P(A,B)} = \frac{P(A|B)P(B)}{\sum_B P(A|B)P(B)}$$

Based on the probabilities given in the exercise, we have for the numerator that

$$P(A|B)P(B) = \begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.1 & 0.7 & 0.6 \\ a_2 & 0.9 & 0.3 & 0.4 \\ \hline\end{array} \cdot (0.1, 0.5, 0.4) = \begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.01 & 0.35 & 0.24 \\ a_2 & 0.09 & 0.15 & 0.16 \\ \hline\end{array}$$

Note that

- $P(A|B)P(B)$ is also equal to $P(A,B)$ (an instance of the *chain rule*)!
- the multiplications are done component-wise so that we only multiply entries that match wrt. the state labels.

For the denominator, we can use the intermediate result ($P(A,B)$) that we just calculated above:

$$P(A) = \sum_B P(A,B) = \sum_B \left(\begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.01 & 0.35 & 0.24 \\ a_2 & 0.09 & 0.15 & 0.16 \\ \hline\end{array} \right) = (0.6, 0.4).$$

Plugging these results into Bayes rule we get:

$$\begin{aligned} P(B|A) &= \begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.01 & 0.35 & 0.24 \\ a_2 & 0.09 & 0.15 & 0.16 \\ \hline\end{array} \Big/ (0.6, 0.4) \\ &= \begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.01/0.6 & 0.35/0.6 & 0.24/0.6 \\ a_2 & 0.09/0.4 & 0.15/0.4 & 0.16/0.4 \\ \hline\end{array} \\ &= \begin{array}{|c|c|c|c|}\hline & b_1 & b_2 & b_3 \\ \hline a_1 & 0.0167 & 0.5833 & 0.4000 \\ a_2 & 0.2250 & 0.3750 & 0.4000 \\ \hline\end{array} \end{aligned}$$

Exercise 9 :

A routine DNA test is performed on a person (this exercise is set in the not too distant future!). The test T

gives a positive result for a rare genetic mutation M linked to Alzheimer's disease. The mutation is present in only 1 in a million people. The test is 99.99% accurate, i.e. it will give a wrong result in 1 out of 10000 tests performed. Should the person be worried, i.e., what is the probability that the person has the mutation given that the test showed a positive result?

Hint: This is partly a modeling exercises and partly a calculation exercise. First you need to formalize the problem:

- What are the relevant variables and what states do they have?
- Based on the description above, what probability distributions can you infer for the variables?

Based on this formalization, you need to find the rules required to answer the question about the probability of a mutation given a positive test result.

Solution: We analyze the problem using the two random variables $\mathit{Test} \in \{\mathit{positive}, \mathit{negative}\}$, $\mathit{Mutation} \in \{\mathit{present}, \mathit{absent}\}$. What our patient should be interested in is the probability of having the mutation, given everything he/she knows, i.e. the fact that there was a positive test result. Thus, we need to compute

$$P(\mathit{Mutation} = \mathit{present} \mid \mathit{Test} = \mathit{positive}).$$

According to Bayes rule, this is equal to

$$P(\mathit{Test} = \mathit{positive} \mid \mathit{Mutation} = \mathit{present}) \frac{P(\mathit{Mutation} = \mathit{present})}{P(\mathit{Test} = \mathit{positive})}$$

Of the probabilities on the right, we have $P(\mathit{Test} = \mathit{positive} \mid \mathit{Mutation} = \mathit{present}) = 0.9999$, and $P(\mathit{Mutation} = \mathit{present}) = 0.000001$. We still need $P(\mathit{Test} = \mathit{positive})$. This can be computed as

$$\begin{aligned}
P(\mathit{Test} = \mathit{positive}) &= P(\mathit{Test} = \mathit{positive}, \mathit{Mutation} = \mathit{present}) \\
&\quad + P(\mathit{Test} = \mathit{positive}, \mathit{Mutation} = \mathit{absent}) \\
&= P(\mathit{Test} = \mathit{positive} \mid \mathit{Mutation} = \mathit{present}) P(\mathit{Mutation} = \mathit{present}) \\
&\quad + P(\mathit{Test} = \mathit{positive} \mid \mathit{Mutation} = \mathit{absent}) P(\mathit{Mutation} = \mathit{absent}) \\
&= 0.9999 \cdot 0.000001 + 0.0001 \cdot 0.999999 \sim 0.0001
\end{aligned}$$

We now get

$$P(\mathit{Mutation} = \mathit{present} \mid \mathit{Test} = \mathit{positive}) = 0.9999 \cdot \frac{0.000001}{0.0001} = 0.009999.$$

Thus, there is only about a 1% chance that the patient has the mutation.

Exercise 10 :

Assume it is your responsibility to monitor volcanic eruptions. You receive data from two different stations (seismometers), S_1 and S_2. Each S_i is modeled as a Boolean variable where "true" stands for "I detected an eruption" and "false" stands for "I did not detect an eruption". The seismometers are not fully reliable, however; they may not detect an eruption even though there was one, and they may mistake an earthquake for an eruption of a volcano. We model this situation with two additional Boolean variables: V for volcanic eruption, and E for Earthquake.

Use the algorithm from the lecture to construct a Bayesian network for these 4 variables. Do so for the following two variable orders:

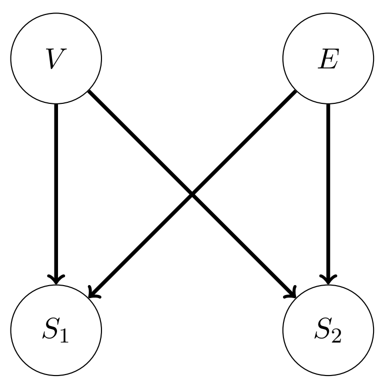
(a) $X_1 = V$, $X_2 = E$, $X_3 = S_1$, $X_4 = S_2$.

(b) $X_1 = S_1$, $X_2 = S_2$, $X_3 = E$, $X_4 = V$.

For each of these orders, draw the resulting Bayesian network. Justify your design, i.e., for each variable X_i added to the network explain why the set of parents you give X_i are needed, and why they are sufficient.

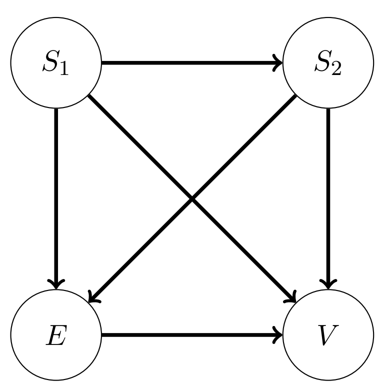
Solution:

(a) With this variable order, we get the following network:



$X_2 = E$ does not need $X_1 = V$ as a parent because Earthquakes are independent from volcano tests. $x_3 = S_1$ needs both $X_1 = V$ and $X_2 = E$ as parents because each of these may influence the measurement; same for $X_4 = S_2$, i.e., here we also need the parents $X_1 = V$ and $X_2 = E$. However, given the values of V and E, the measurements of $X_3 = S_1$ and $x_4 = S_2$ are independent. So $X_4 = S_2$ does not require the parent $X_3 = S_1$.

(b) With this variable order, we get the following network:



	$A = yes$	$A = no$
$T = yes$	0.99	0.001
$T = no$	0.01	0.999

Table 4: Table for Exercise 11. Conditional probabilities $P(T \mid A)$ characterizing test T for A.

$X_2 = S_2$ needs $X_1 = S_1$ as a parent because, if S_1 detects a seismic phenomenon, then chances are higher S_2 will detect one as well. $X_3 = E$ needs each of $X_1 = S_1$ and $X_2 = S_2$ as parents because, if a station detects a seismic phenomenon, then chances are higher there was an earthquake; same for $X_4 = V$, i.e., here we also need the parents $X_1 = S_1$ and $X_2 = S_2$ because measurements indicate volcano tests as well. Finally, say we already know that S_1 and S_2 are true; then the value of E still has an influence on the value of V: If there was an earthquake, then there is a chance that the seismic measurements were caused by the earthquake rather than a volcano test. Thus V is *not* conditionally independent of E given S_1 and S_2, and we need $X_3 = E$ as a parent of $X_4 = V$ as well.

Exercise 11 :

Table 4 describes a test T for an event A. The number 0.01 is the frequency of *false negatives*, and the number 0.001 is the frequency of *false positives*.

(*i*) The police can order a blood test on drivers under the suspicion of having consumed too much alcohol. The test has the above characteristics. Experience says that 20% of the drivers under suspicion do in fact drive with too much alcohol in their blood. A suspicious driver has a positive blood test. What is the probability that the driver is guilty of driving under the influence of alcohol?

(*ii*) The police block a road, take blood samples of all drivers, and use the same test. It is estimated that one out of 1,000 drivers have too much alcohol in their blood. A driver has a positive test result. What is the probability that the driver is guilty of driving under the influence of alcohol?

Solution: We use Bayes rule to calculate the probabilities.

$$\begin{aligned} P(A = y \mid T = p) &= \frac{P(A = y, T = p)}{P(T = p)} \\ &= \frac{P(T = p \mid A = y)P(A = y)}{P(T = p)} \end{aligned}$$

From the problem specification, we have the numbers to put into the numerator, but we still need to calculate the denominator:

$$\begin{aligned} P(T = p) &= P(T = p, A = y) + P(T = p, A = n) \\ &= P(T = p \mid A = y)P(A = y) + P(T = p \mid A = n)P(A = n) \end{aligned}$$

	b_1	b_2
a_1	$(0.006, 0.054)$	$(0.048, 0.432)$
a_2	$(0.014, 0.126)$	$(0.032, 0.288)$

Table 5: $P(A, B, C)$ for Exercise 12.

By plugging this into the previous expression we get:

$$\begin{aligned} P(A = y \,|\, T = p) &= \frac{P(T = p \,|\, A = y)P(A = y)}{P(T = p \,|\, A = y)P(A = y) + P(T = p \,|\, A = n)P(A = n)} \\ &= \frac{0.99 \cdot 0.2}{0.99 \cdot 0.2 + 0.001 \cdot 0.8} \\ &= 0.996 \end{aligned}$$

In the second part of the exercise we change the prior probability $P(A)$ for A and update the calculations using the new probabilities. This gives $P(A = y) = 0.498$. Notice how the change in prior distribution affected the posterior distribution for A; the accuracy of a test should always be considered relative to the frequency of the event which you are trying to predict.

Exercise 12 :

In Table 5, a joint probability table for the binary variables A, B, and C is given.

- Calculate $P(B, C)$ and $P(B)$.
- Are A and C independent given B?

Solution:

For the first part of the exercise, consider calculating $P(B, c_1)$. To do that we marginalize out A, hence

$$P(B, c_1) = (0.006 + 0.014, 0.048 + 0.032) = (0.02, 0.08).$$

The remaining probabilities are calculated similarly, and we get:

(i) $P(B, c_1) = (0.02, 0.08), P(B, c_2) = (0.18, 0.72), P(B) = (0.2, 0.8)$

For the second part of the exercise we can check that

$$P(A|C = c_1, B) = P(A|C = c_2, B)$$

to verify that A and C are independent given B. These probabilities can be calculated as

$$P(A|B, C) = \frac{P(A, B, C)}{P(B, C)},$$

where the denominator was calculated in the first part of the exercise. When doing the calculations we get

(ii) $P(A|b_1, c_1) = (0.3, 0.7) = P(A|b_1, c_2), P(A|b_2, c_1) = (0.6, 0.4) = P(A|b_2, c_2)$,

hence A and C are independent given B.